Ekarsi Lodh

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Education

2025 - Present — MSc in Bioinformatics, **University of Birmingham, United Kingdom**
Thesis: *Epigenetic Age Acceleration as a Marker of Early-Onset Breast Cancer: A Multi-Cohort DNA Methylation Clock Analysis*, under the supervision of ***Dr Archana Sharma-Oates***, University of Birmingham, UK

2020 - 2025 — BTech in Computer Science and Engineering, **Techno Main Salt Lake, India** (CGPA: 8.13/10.0)

Research Experience

Research Associate (Mar 2026 – Present)
HealthTech AI Hub, University of Birmingham, UK
Under the supervision of ***Dr Mamunur Rashid***

- Conducting research on machine learning and AI applications for Electronic Health Record (EHR) data.
- Training and fine-tuning the **Internist-7B clinical large language model** and **Gemma-9B model** on multiple EHR datasets to predict patient mortality and hospital readmission risk.
- Implementing parameter-efficient fine-tuning methods (QLoRA/PEFT) and benchmarking against transformer and classical machine learning models.

MSc Thesis Project (Nov 2025 – present)

- Currently working under the supervision of ***Dr Archana Sharma-Oates***, Assistant Professor in Bioinformatics, School of Biosciences, University of Birmingham, UK
  Research Topic: Epigenetic Age Acceleration and Earlier Breast Cancer Onset in UK-Resident South Asian Women: A multi-cohort DNA methylation clock study. The project investigates ethnicity-associated differences in epigenetic ageing using blood-derived DNA methylation data, integrating multiple generations of epigenetic clocks (Horvath, Hannum, PhenoAge, GrimAge, and DunedinPACE). The work focuses on robust preprocessing, cell composition adjustment, and multi-clock triangulation to assess associations with breast cancer risk and age at diagnosis, with an emphasis on underrepresented populations and reproducible statistical and machine learning workflows implemented in R and Python.

Group Research Project (Oct 2025 - present)

- Currently working under the supervision of ***Dr Lindsey Compton***, Associate Professor in Genetics, School of Biosciences, University of Birmingham, UK
  Research Topic: Optimising GWAS pipelines to uncover genetic loci underlying canopy architecture and yield (complex traits) in autotetraploid potato using SNP and phenotyping datasets, enabling enhanced precision breeding for climate-resilient crop improvement. The work integrates population structure analysis (DAPC and STRUCTURE), environmental correction using Best Linear Unbiased Estimates (BLUEs), and comparative statistical models to improve trait–locus associations. Gaining hands-on experience in developing a robust polyploid GWAS workflow using R and Linux.

Research Internship (Feb 2023 - Sept 2025)

- Pursued under the supervision of ***Dr Tapan Chowdhury***, Head and Associate Professor, Department of Computer Science and Engineering, Techno Main Salt Lake, Kolkata, India
- **Research Domain:** Signaling Pathway Analysis, Molecular Docking, Drug Repurposing, Deep Learning for Protein-Ligand Binding Affinity Prediction, Sequence Alignment Algorithms

Teaching & Mentoring Experience

Undergraduate Research Project Co-Supervisor (Aug 2025 - Present)
Techno Main Salt Lake, India

- Co-supervising a group of B.Tech final-year students under the supervision of ***Dr Tapan Chowdhury***, Head and Associate Professor, Department of Computer Science and Engineering, Techno Main Salt Lake, Kolkata, India
- Conducting regular remote mentoring sessions from the UK to guide literature review, research design, and computational methodology
- Guided students in conducting a **miRNA–mRNA interaction network analysis for Alzheimer's disease biomarker discovery**
- Mentored students in scientific writing and manuscript preparation; the work resulted in a **student-led bioRxiv preprint currently under review at *Journal of Molecular Neuroscience***

Publications-Journal

[1] **E. Lodh**, S. Majumder, T. Chowdhury, and M. De, "NetPolicy-RL: Network-Informed Offline Reinforcement Learning for Pharmacogenomic Drug prioritization," *Journal of Computer-Aided Molecular Design*, vol. 40, no. 140, p. 24, Jun. 2026, doi: 10.1007/s10822-026-00823-4.

[2] S. Majumder, **E. Lodh**, T. De, and T. Chowdhury "In silico Characterization of Melanoma-derived Mouse lncRNA

Gm26982 and Comparative Analyses with Human Ortholog," *Molecular Genetics and Genomics*, vol. 301, no. 103, p. 25, Apr. 2026, doi: 10.1007/s00438-026-02423-1.

[3] **E. Lodh**, T. Chowdhury, and S. Majumder, "BioAlignNet: A GPU-Accelerated Framework for Efficient Global Sequence Alignment Across Genomics and Proteomics Scales," *Computational Intelligence*, vol. 42, no. 3, p. e70232, Apr. 2026, doi: 10.1111/coin.70232.

[4] **E. Lodh**, T. Chowdhury, A. Shaw, D. Bedajna, S. Bera, S. Majumder, and M. De "A Stacked Ensemble Meta-Classifier Framework for PAM50 Breast Cancer Subtype Prediction from Gene Expression Signatures," *Network Modeling Analysis in Health Informatics and Bioinformatics*, vol. 15, no. 78, p. 31, Mar. 2026. doi: 10.1007/s13721-026-00755-x.

[5] **E. Lodh**, S. Majumder, T. Chowdhury, and M. De, "RLBindDeep: A ResNet-LSTM based novel framework for protein–ligand binding affinity prediction," *Journal of Molecular Graphics and Modelling*, vol. 144, no. 109282, p. 24, Jan. 2026, doi: 10.1016/j.jmgm.2026.109282

[6] T. Chowdhury, S. Basu, A. Roy, V. Sarkar, K. Mondal, **E. Lodh**, and M. De, "Graph-theoretic and emotional analysis of character dynamics in novel-to-film adaptations," *Journal of Computational Social Science*, vol. 9, no. 18, p. 23, Jan. 2026. doi: 10.1007/s42001-025-00451-2

[7] S. Majumder, **E. Lodh**, and T. Chowdhury, "Implications of trinodal inhibitions and drug repurposing in MAPK pathway: A putative remedy for breast cancer," *Computational Biology and Chemistry*, vol. 113, no. 108255, p. 22, Oct. 2024, doi: 10.1016/j.compbiolchem.2024.108255.

Publications-Preprint

[1] A. Ray, K. Agarwal, S. Jha, A. M. Singh, **S. Majumder**, E. Lodh, and T. Chowdhury, "miRNA-–mRNA interaction network analysis in Alzheimer's disease for biomarker discovery," *bioRxiv (Cold Spring Harbor Laboratory)*, Jan. 2026. doi: 10.64898/2026.01.28.702349 (Manuscript under revision at Journal of Molecular Neuroscience).

[2] T. Chowdhury, S. Majumder, **E. Lodh**, A. Maitra, A. Agarwal, A. Sur, and S. Sarkar, "Multiple Fault Analysis and Drug Therapy on Signaling Pathways Using Dynamic Bayesian Network-based Model," *bioRxiv (Cold Spring Harbor Laboratory)*, Jun. 2026. doi: 10.64898/2026.06.11.731601 (Manuscript submitted to The FEBS Journal).

Publications-Conference

[1] **E. Lodh**, S. Majumder and T. Chowdhury, "CGDeepAff: Deep Learning-Based Approach for Protein-Ligand Binding Affinity Estimation Using CNN-GRU," *2025 8th International Conference on Electronics, Materials Engineering & Nano-Technology (IEMENTech)*, Kolkata, India, 2025, pp. 1-6, doi: 10.1109/IEMENTech65115.2025.10959578. (**Presenter**)

[2] T. Chowdhury, U. Das, A. Makur, A. Sinha, D. Jana, A. Das, and **E. Lodh**, "Ensemble Clustering on Big Data," *2025 8th International Conference on Electronics, Materials Engineering & Nano-Technology (IEMENTech)*, Kolkata, India, 2025, pp. 1-6, doi: 10.1109/IEMENTech65115.2025.10959554. (**Presenter**)

[3] M. De, R. L. Chhetri, M. Konar, A. N. Joardar, A. Jain, **E. Lodh** and T. Chowdhury, "A Deep Learning-Based Method for the Categorization of Different Skin Diseases," W. Ding, A. Chakrabarti, M. Chakraborty, S. Chakraborty, (eds) *Proceedings of the Second International Conference on Advanced Computing and Systems, AdComSys 2025, Lecture Notes in Networks and Systems*, vol 1887, pp. 40-56 Springer, Cham, doi: 10.1007/978-3-032-20253-6_4. (**Presenter**)

[4] **E. Lodh**, S. Majumder, T. De, T. Chowdhury, and M. De, "GeneDeepNet: A Differential Expression and Deep Learning Based Novel Framework for Detecting Invasive Breast Cancer Subtypes," *4th International Conference on Machine Learning and Data Engineering (ICMLDE-2025), Procedia Computer Science*, 2025 (Accepted, In Press). (**Presenter**)

Publications-Book Chapter (Invited)

[1] S. Majumder*, **E. Lodh*** (joint first authors), and S. Datta, "Explainable AI in Redox Medicine: Challenges and Opportunities," in *Oxidative Stress–Mediated Disorders: Advances in Diagnosis and Therapy with AI and Machine Learning*, edited by S. Roy, D. Ghosh, P. Kalita, and P. H. Guzzi, Springer Nature, 2026. (Invited Chapter, In Preparation).

Projects

Clinical Mortality Risk Prediction using Fine-Tuned Clinical LLMs (2026)
Machine Learning Module Project; Healthcare AI & Large Language Models
- Developed in-hospital mortality prediction models using structured EHR data inspired by MIMIC clinical datasets.
- Fine-tuned **Internist-7B** using **QLoRA (4-bit PEFT)** for efficient clinical outcome prediction.
- Fully fine-tuned **BioClinicalBERT** and benchmarked against XGBoost, LSTM, and transformer models.
- Evaluated performance using **AUC-ROC, Precision, Recall, and F1-score**; GitHub release planned.

Streamlit-Based Trio VCF Analyzer for Structural Variant Interpretation (Launch App, GitHub Repo) (2025)
Based on Genomics & NGS Module Project · Interactive Web App
- Developed an interactive Streamlit app that ingests VCF files from trio-based structural variant calling and presents a complete summary of SV types, genomic locations, and functional context.
- Implements trio role inference (child/parents), sex inference based on X-chromosome heterozygosity, and detection of de novo mutations via Mendelian consistency checks.

Delly-VCF-Trio-Analyzer: Command-Line SV Interpretation Pipeline (GitHub Repo) (2025)
Genomics & NGS Module Project; Reproducible Analysis Workflow

- End-to-end command-line pipeline for parsing, summarising, and prioritising structural variants from Delly-generated trio VCF files, including QC, de novo calling, annotating, tiering pathogenic variants, and plotting.
- Designed for reproducible workflows with clearly structured output tables, text reports, and visualisations that can be integrated into downstream clinical or research pipelines.

Professional Memberships

IEEE (Institute of Electrical and Electronics Engineers) Member Feb 2025 - present

– Member of IEEE Computational Intelligence Society and IEEE Engineering in Medicine and Biology Society

Leadership

2025 - present	Postgraduate Student Ambassador, **University of Birmingham**
2025 - present	Student Rep, MSc Bioinformatics FT (2025-26 Batch), **University of Birmingham**
2024 - 2025	Convenor of Department of Computer Science and Engineering, **Techno Main Salt Lake**
2023 - 2025	Class Representative, Department of Computer Science and Engineering, **Techno Main Salt Lake**

Certifications

Biology Meets Programming: Bioinformatics for Beginners from University of California San Diego, Coursera	(2025)
Python for Genomic Data Science from Johns Hopkins University, Coursera	(2024)
Introduction to Machine Learning from Duke University, Coursera	(2023)
Introduction to Python Programming from University of Pennsylvania, Coursera	(2022)

Language Qualifications

IELTS, Overall Band Score: 8.0 (Listening: 8.5, Reading: 9.0, Writing: 8.0, Speaking: 6.5), CEFR level: C1	(2025)
Goethe Start Deutsch A1, Goethe-Institut, Score: 71/100, Befriedigend	(2024)

Skills

Programming Languages
Python, R, Shell Scripting, Awk, Linux, C

Machine Learning & AI

- Deep Learning and Transformer Models (PyTorch, HuggingFace)
- Large Language Model Fine-Tuning (**QLoRA, PEFT, 4-bit Quantisation**)
- Clinical NLP and Healthcare Predictive Modelling
- BioClinicalBERT and Domain-Specific Transformer Adaptation
- Sequential Modelling (LSTM) and Model Evaluation (AUC-ROC, F1-score)

Bioinformatics Tools

- Proficiency in molecular docking and real-time simulation tools (e.g., **UCSF Chimera, PyRx, AutoDock Vina 1.2.5, ClusPro 2.0, Discovery Studio (BIOVIA)**)
- Familiar with in silico ncRNA chracterization tools (e.g., **RNAfold, LncRRISearch, LncBase v.3, lncLocator, iLOC-LncRNA(2.0), CPC2, RNAinter v4.0, NPInter v5.0**)
- Familiar with primer tools (e.g., **NEBcutter V2.0, Primer3 Input**)
- Done working with plant regulome analysis tools (e.g., **PlantPAN 4.0, Phytozome v13**)

High-Performance & Large-Scale Computing

- Gained experience in running large-scale genomics workflows in BlueBEAR HPC (University of Birmingham)
- GPU-accelerated deep learning and LLM fine-tuning on BlueBEAR HPC.

Digital Skills
Microsoft Office tools, Canva, DATAtab, Cytoscape

Editing Softwares
Adobe Photoshop, Adobe Premier, Adobe Audition

Relevant Experience

AI Hackathon Volunteer, HealthTech AI Hub & University of Birmingham (Mar 2026)

- Supported the organisation and delivery of a large-scale international healthcare AI hackathon in a fast-paced environment.
- Assisted participants by resolving queries, providing guidance, and enhancing overall participant experience.
- Contributed to the technical evaluation of an NLP model using a Bengali dataset, reviewing the accuracy of translating medical check-up data into English and ensuring clinical relevance.
- Contributed to event coordination, including setup, logistics, and real-time operational support.
- Conducted interviews with student teams and industry partners, gaining insights into solution design and problem-solving approaches.
- Supported talent matching activities between participants and industry stakeholders.
- Captured key event moments and supported documentation of major activities.

Competitions

Internal Hackathon'23 Sept 2023

- Intra-college hackathon for finals of SIH-2023
- Organised by Techno Main Salt Lake
- **Position:** Qualified to the finals

LOKNITI, Policy Conclave 2021 Apr 2021

- Organised by Public Policy & Opinion Cell, IIT Kanpur
- **Project:** ACCEPTaB (Activated, controllable, cost-effective, eco-friendly, perennial, thermo-absorbent bio-generator
- **Position:** 1st

Extracurriculum

Participation in the cultural programme at the Jahresabschlussfeier at Goethe-Institut Kolkata (2024)
Organised Hasta-La-Vista, Dept. of CSE, Techno Main Salt Lake (2023)
Participation in Photography event, by Calcutta Youth Meet Chapter 5 (2021)
1st in TRIFECTA, a Fandom Quiz - by Literary Club of Techno Main Salt Lake (2021)
Participation in SIYAAHI (Bengali), a Writing Competition, by Heramba Chandra College (2021)
Passed Sangeet Visharad Part 1 (4th Year) in Vocal-Classical, from Pracheen Kala Kendra (2016)